Machine Learning of 3D Scene Geometry
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MASTER'S THESIS

Foundation-Model-Guided Zero-Shot Synthesis of Human-Scene Interaction

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Start Date:	19.01.2026
Submission Date:	20.07.2026

Declaration

I declare that this thesis has been completed independently and that no sources or aids other than those listed have been used.

Munich, 20.07.2026

Place, Date

Signature

Abstract

Generating a human that interacts convincingly with a real 3D scene is difficult because the input is semantic while the output must satisfy geometry. A hand must meet the correct object surface, a foot must land where the body is supported, and the body must not float above the floor or pass through nearby furniture. What separates a believable interaction from a merely photogenic one is correct contact with the actual reconstructed geometry, and the hardest part of establishing that contact is determining, precisely and for each participating body part, which region of the scene surface the interaction requires it to touch.

Existing zero-shot approaches leave this scene-side contact only partially resolved: they read contact implicitly from a generated image, or commit to a functional scene element in advance, which under-determines the contact on large surfaces. This thesis addresses that gap with a zero-shot pipeline that synthesizes a single static human–scene interaction, placed in the coordinate frame of a reconstructed scene, from a natural-language instruction, a single posed RGB view, and the reconstructed ScanNet++ scene mesh, using only pretrained foundation models together with explicit camera geometry and optimization, with no interaction-specific training.

The central idea is to use foundation models not to produce the final result but to *localize* where the interaction touches the scene. The core contribution is an agentic contact-estimation procedure: an image generation model proposes body-part-specific contact regions, which are extracted and composited onto the untouched posed view so that they stay in exact correspondence with the calibrated camera, and a vision-language evaluator inspects each result against the intended interaction and issues corrections in a closed loop until the contacts are accepted. This verify-and-correct loop turns an unreliable single generation into precise, per-body-part contact regions, which are projected onto the mesh as 3D targets. Around this contribution, a lightweight interaction graph fixes the coarse contact structure, a monocular recovery model initializes the body, and a scene-aware optimization refines it to meet the contact targets while avoiding penetration and remaining physically plausible.

The method is evaluated on reconstructed ScanNet++ scenes using metrics for physical contact, scene penetration, interaction diversity, image–text consistency, and vision-language

verification of the rendered interaction. Together, these results show that grounding foundation-model interaction knowledge on explicit, verified contacts yields geometrically plausible 3D human–scene interactions from text, without task-specific training.

Acknowledgments

Thank advisors, supervisors, collaborators, funding sources, friends, and family here. Keep this section personal and direct.

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List of Abbreviations

2D	Two-Dimensional
3D	Three-Dimensional
4D	Four-Dimensional (spatial and temporal)
BVH	Bounding Volume Hierarchy
HOI	Human-Object Interaction
HSI	Human-Scene Interaction
LLM	Large Language Model
RGB	Red, Green, Blue
RGB-D	Red, Green, Blue, and Depth
SDF	Signed Distance Field
SIG	Scene Interaction Graph
SMPL	Skinned Multi-Person Linear (model)
SMPL-X	SMPL eXpressive
VLM	Vision-Language Model

Chapter 1

Introduction

1.1 Motivation

Advances in 3D scanning and reconstruction have made detailed digital replicas of real environments widely available. Datasets such as ScanNet++ [1] provide high-fidelity reconstructions of indoor spaces, and modern reconstruction pipelines can turn ordinary captures into accurate 3D scene meshes. Yet these reconstructed environments are, almost without exception, empty: they faithfully capture rooms, furniture, and objects, but not the people who give a space its purpose. A gym is reconstructed without anyone running on its treadmill; a kitchen is reconstructed without anyone opening its appliances. Bridging this gap, populating reconstructed scenes with believable humans that genuinely *interact* with their surroundings, is the problem this thesis addresses (figure 1.1).

The ability to place plausible interacting humans into arbitrary 3D scenes is valuable across several domains. In embodied AI and robotics, agents must anticipate how people occupy and act within an environment in order to navigate and assist safely, and synthetic humans placed in reconstructed scenes provide training and evaluation data for this reasoning. In virtual and augmented reality, animation, and games, populating captured or designed environments with lifelike interacting characters is a recurring need. And across computer vision more broadly, human–scene interactions rendered in known 3D scenes offer a scalable source of annotated data, with ground-truth body pose and contact, for tasks such as human pose and contact estimation, where real annotated data is expensive to capture. In each case, the requirement is not merely a human placed somewhere in the scene, but a human placed *in interaction with it*: contacting the right surfaces, in a pose that the interaction demands.

What makes this requirement demanding is that plausibility is fundamentally a three-dimensional, physical property. A body may appear convincing from one viewpoint yet be physically wrong in the scene: floating above the floor, intersecting furniture, or reaching

Teaser placeholder (treadmill running example)

Input: posed ScanNet++ view I (empty treadmill) + instruction

"a person running on the treadmill, one foot landing on the moving belt, the other leg lifted behind, and both hands lightly holding the side handles"

$\Longrightarrow$

Output: SMPL-X body in correct contact, placed in the scene coordinate frame

Figure 1.1: Given a single posed RGB view of a reconstructed scene and a natural-language instruction, the proposed method synthesizes a static human–scene interaction: a single SMPL-X body placed in the scene's coordinate frame, in correct physical contact with the geometry. Here, the empty treadmill on the left is populated with a person running on it, whose hands meet the side rails and whose foot lands on the belt.

toward a surface it never actually meets. What distinguishes a believable interaction from a merely photogenic one is correct *contact* with the real scene geometry, the hand truly meeting the handle, the foot truly landing on the surface it pushes against. Generating such interactions therefore cannot be treated as an image problem alone; it requires reasoning about where, on the actual reconstructed geometry, the body and the scene should meet. Establishing that contact, precisely and for each participating body part, is the central challenge that this thesis takes up.

1.2 Problem Statement

This thesis studies a specific formulation of human–scene interaction synthesis, chosen to isolate the contact challenge described above. The method takes three inputs: a natural-language instruction describing the desired interaction, a single posed RGB view of a reconstructed scene, with known camera intrinsics and extrinsics, and the reconstructed scene mesh. From these, it produces a single SMPL-X body placed in the coordinate frame of the reconstructed scene, positioned and posed so that it performs the described interaction in correct physical contact with the scene geometry.

Two aspects of this formulation are deliberate. First, the target is a single *static interaction state* rather than a motion sequence: one body configuration that realizes the requested interaction, such as the moment a runner is in contact with a treadmill, rather than an animation of the activity over time. A static state is not a simplified motion; it is a distinct target in which the body must be resolved into one specific configuration in metric contact with the scene, with no trajectory over which small errors can be absorbed. Second, the method is *zero-shot*: it uses only pretrained foundation models together with explicit camera geometry and optimization, and requires no training on paired human–scene interaction data. This is what allows it to operate on arbitrary reconstructed scenes and open-vocabulary instructions, rather than being confined to the environments and interactions of a capture dataset.

The difficulty of this problem lies in the mismatch between the spaces in which its inputs and its output live. The instruction is linguistic, the view is a two-dimensional image, and the scene is a three-dimensional mesh; the output must be a body that is correct in the metric 3D frame of that mesh. Foundation models can readily imagine what the interaction looks like as an image, a person plausibly running on the treadmill, but an image does not, on its own, specify where in the reconstructed scene the runner's hands and feet make contact. A body recovered from such an image is expressed in image space and carries no guarantee of metric consistency with the scene: it may look correct in the view yet float above the belt or reach past the handles in 3D. Turning the interaction knowledge of foundation models into a body that is physically correct on the reconstructed mesh therefore requires an explicit bridge from image evidence to scene geometry, and the load-bearing part of that bridge is determining, precisely and for each body part involved, which region of the scene surface the interaction requires it to contact.

1.3 Research Gap

Existing approaches to human–scene interaction synthesis do not provide the capability this problem requires. Methods that model contact do so on the *body side*, predicting which vertices of the human body are in contact, and they learn this from captured paired data, which ties them to the scenes and interactions of their training sets and does not transfer to arbitrary reconstructed environments. Language- and semantics-driven methods add control over the interaction, but that control operates at the level of the object and the action: they determine which object is involved and what is done with it, not the precise region of its surface that each body part meets. More recent zero-shot methods, which distil interaction knowledge from foundation models without any interaction-specific training, come closest, but they resolve the scene side of the contact only partially. GenZI [2] treats the generated interaction image as an implicit whole and fits a body to its overall appearance, without extracting explicit, verified contact regions. FunHSI [3] grounds contact functionally, but commits to functional scene elements symbolically and in advance, which leaves the contact under-determined when the element is large: identifying a tabletop or a rail does not say where along that surface the contact falls. A full account of this prior work is given in chapter 2.

Taken together, these lines of work leave one capability unaddressed: the recovery of *precise, verified, per-body-part contact regions on the surface of an object embedded in a reconstructed scene, in a zero-shot manner.* Each qualifier matters. The contact must be *precise*, resolving the specific surface patch rather than the object or element as a whole; *verified*, so that a wrong or missing contact can be detected and corrected rather than propagated; *per body part*, pairing each named part with its own scene-side region; grounded on a *reconstructed scene* rather than an isolated object; and obtained *zero-shot*, without interaction-specific training. It is the conjunction of these properties, not any single one, that no prior method delivers, and closing this gap is the objective of this thesis.

1.4 Research Question

The gap identified above leads to the central research question of this thesis:

> *Can a zero-shot pipeline, using only pretrained foundation models and explicit camera geometry, recover precise, verified, per-body-part contact regions on the surface of a reconstructed scene, and use them to synthesize a plausible static human–scene interaction placed in the scene's coordinate frame?*

This question is examined through three more specific sub-questions, which together structure the method and its evaluation:

1. **Localization.** Can an image generation model be used to *localize* fine-grained, body-part-specific contact regions on a scene surface, reliably enough that they can be projected into 3D and used as optimization targets?
2. **Verification.** Does a closed-loop, agentic verify-and-correct procedure produce more reliable contact regions than a single generation, by detecting and correcting contacts that are missing, misplaced, or on the wrong surface?
3. **Grounding.** Does grounding the interaction on the reconstructed scene geometry, through these localized contacts, yield bodies that are more physically plausible than those obtained by fitting to image evidence alone?

The first sub-question concerns the feasibility of the core idea, using generation to localize rather than to render the final result; the second isolates the contribution of the agentic loop that is central to this thesis; and the third tests whether explicit scene-side grounding delivers the physical plausibility that motivates the whole approach. These sub-questions are revisited in the evaluation of chapter 4 and answered in the conclusion.

1.5 Approach

The key idea of this thesis is to use foundation models not to produce the final interaction directly, but to *localize* where the interaction touches the scene. Rather than generating an image and treating it as the answer, the method uses image generation as a way to reason about contact: it inpaints a human performing the interaction into the scene, uses that rendered interaction to mark where each body part meets the scene surface, and then grounds those marked regions on the reconstructed mesh through the calibrated camera. Because the contact regions are localized in the same posed view whose geometry is known, they can be projected into 3D exactly, giving explicit contact targets that a body can be optimized to satisfy. Generation supplies the interaction knowledge; camera geometry and optimization turn it into a physically correct body.

Making these localized contacts reliable is the role of the method's central contribution: an *agentic contact estimation* procedure. A single generated contact annotation cannot be trusted, it may place a contact on the wrong surface, omit one the interaction requires, or draw one of the wrong size, so the method does not consume it in one shot. Instead, an image generation model proposes body-part-specific contact regions, which are extracted and composited back onto the untouched posed view so that they remain in exact correspondence with the calibrated camera and the reconstructed mesh, and a vision-language evaluator then inspects each result against the intended interaction and, when it finds an error, issues a correction for another round. This closed verify-and-correct loop converts an unreliable single generation into a dependable set of precise, per-body-part contact regions. Because the regions are localized visually rather than inferred from the identity of a named element, the procedure resolves contact on small and large functional surfaces alike.

These ideas are realized in a five-module pipeline. A Scene Interaction Graph first converts the instruction into a coarse, symbolic specification of which body parts contact the target object or the floor. A human frame is then generated, inpainting a single person performing the interaction into the posed view. From this frame, two branches run in parallel: agentic contact estimation localizes and verifies the scene-side contact regions and projects them to the mesh, while a monocular recovery model initializes a plausible SMPL-X body from the same frame. A final scene-aware optimization reconciles the two, refining the body so that its contact regions meet the projected targets while it avoids penetrating the scene and stays close to the initialized pose. The symbolic graph that seeds this process is deliberately lightweight: it fixes only the coarse contact structure and leaves the precise surface region to be recovered visually and verified downstream. The complete method is developed in chapter 3.

Chapter 2

Related Work

The task addressed in this thesis is to synthesize a plausible static human–scene interaction, in the coordinate frame of a reconstructed scene, from a natural-language instruction and a single posed view, without any interaction-specific training. This chapter reviews the work that leads up to that task and isolates the specific capability it requires. The review is organized around a single question that recurs at every stage: not merely *whether* a body contacts the scene, but *which body part contacts where on the scene surface*, and how precisely that correspondence can be recovered. This section begins with the representations that established contact as the basis of interaction; section 2.2 turns to language- and semantics-driven synthesis; section 2.3 examines foundation-model and zero-shot methods; and section 2.4 states the resulting gap and positions the proposed method against it.

2.1 Contact as the Basis of Human–Scene Interaction

A human body placed in a scene looks plausible only when it makes physical sense: feet resting on the floor, a hand meeting a handle, hips settling onto a seat. What separates a convincing interaction from a floating or interpenetrating body is contact, the set of points at which body and scene meet. This observation has made contact the organizing principle of human–scene interaction (HSI) modeling, and the representations developed to capture it form the foundation on which later synthesis methods build.

Early work established contact and proximity as explicit constraints on body placement. PROX [4] showed that a 3D scene can be used to resolve the ambiguity of monocular human pose estimation, by requiring the estimated body to remain in contact with the floor and to avoid penetrating scene geometry. Contact here is a corrective signal: the scene constrains an otherwise underdetermined body. PLACE [5] made the proximity relationship itself the modeled quantity, encoding the distances between the body and nearby scene surfaces so that generated bodies respect the geometry around them. POSA [6] reformulated this relationship in a body-centric way, attaching to each vertex of the SMPL-X body a learned distribution over contact and the semantic category of the surface it should touch. In POSA, the body carries its own per-vertex contact expectations, which are then matched against a scene to place the person.

A parallel line of work sought to detect contact directly from images and to supply the data needed to learn it. RICH [7] introduced a dataset of real indoor and outdoor scenes captured together with ground-truth bodies and dense, vertex-level human–scene contact labels, and trained BSTRO, a transformer that infers per-vertex body contact from a single image. DECO [8] extended dense contact estimation to unconstrained in-the-wild images, predicting which parts of the body are in contact for scenes with no ground-truth geometry. Together, these methods advanced the field from hand-specified contact heuristics toward learned, dense, image-grounded contact prediction.

Two properties of this body of work are important for what follows. First, contact is represented on the *body side*: the modeled quantity is which vertices of the human mesh are in contact, or how close each body vertex lies to the surrounding surface. This is a natural choice, since the body has a fixed topology on which contact can be annotated consistently, and methods such as POSA even resolve it per body part. But it leaves the complementary quantity, *where on the scene surface* the contact occurs, implicit. A body-side contact label states that a palm is touching something; it does not, by itself, identify the region of the object that the palm should meet, nor pair that specific body part with that specific surface region. The relationship that actually determines an interaction, *which body part contacts which scene region*, is therefore left half-specified: the body side is named, and often named per part, but the scene side, and the correspondence between the two, are not. Second, these methods obtain their contact knowledge from *captured paired data*: real scenes recorded together with real bodies and contact annotations. This dependence ties them to the distribution of the capture datasets and to the scenes, objects, and interactions represented there.

For the setting of this thesis, both properties are limiting. The target is a body placed in a *reconstructed* scene the model has never seen, performing an interaction described by *open-vocabulary* language, with the contact grounded on the *scene surface* so that it can drive metric optimization against the reconstructed mesh. Body-side contact representations do not localize the scene-side region, and data-driven contact predictors do not generalize to arbitrary reconstructed environments and instructions without retraining. What is missing is a way to determine, for a given instruction and scene, which body part should contact which region of the scene surface. We refer to this throughout the chapter as *scene-side contact localization*: the recovery, for each participating body part, of the specific region on the reconstructed scene surface that it should contact. The emphasis is twofold: on the *scene side* of the contact, which body-side representations leave implicit, and on the *per-body-part correspondence*, so that each named part, the left hand, the right hand, a supporting foot, is paired with its own surface region rather than the body as a whole being placed against the scene. This is the capability that the representations reviewed here establish the need for but do not provide, and the sections that follow trace how subsequent work approaches it.

2.2 Language- and Semantics-Driven HSI Synthesis

The representations of the previous section make interactions physically plausible, but they do not say *what* interaction to perform. As the field moved from constraining a given body to *synthesizing* interactions, a second requirement emerged: semantic control, the ability to specify the desired interaction through a label or a natural-language description rather than through geometry alone. This section reviews methods that add such control, and shows that while they resolve which object and which action an interaction involves, they stop short of localizing the contact on the object surface.

Semantic control over static interactions. A first group of methods synthesizes a static body performing a described interaction in a given scene. Early affordance-driven work learned where a person could plausibly be placed in an environment, populating scenes with bodies whose poses suit the surrounding geometry [9]. Building on the body-centric contact representation of POSA, COINS [10] introduced compositional semantic control: given an object and an action, such as sitting on a chair or touching a table, it generates a body that performs that specific interaction, and it grounds the interaction on the named object rather than on the scene as a whole. This is a decisive step toward controllable HSI, since the interaction is now specified semantically and attached to a particular object. The control, however, operates at the level of the object and the action: COINS selects *which* object is involved and *what* the interaction is, but the precise region of the object surface that the body contacts is determined implicitly by the learned model, not localized as an explicit target. For a large or extended surface, a tabletop, a countertop, a rail, the object identity does not determine where along that surface the hand should land.

Language-conditioned motion in scenes. A larger body of work considers the dynamic setting, generating human *motion* in a scene from a textual description. HUMANISE [11] paired language-annotated motions with ScanNet scenes to train a language-conditioned model for scene-aware motion synthesis. TeSMo [12] combined a scene-agnostic text-to-motion diffusion model with scene-aware fine-tuning, producing navigation and interaction motions that respect object placement and the ground plane. Cen et al. [13] decomposed the problem into language grounding of the target object followed by object-centric motion generation, using a large language model to identify the object the text refers to. Broader scene-level generative frameworks such as SceneDiffuser [14] unified generation, optimization, and planning of human activity in 3D scenes, and large-scale efforts such as TRUMANS [15] scaled up the paired motion-capture data available for learning dynamic interactions. These methods differ in architecture and in the granularity of their scene reasoning, but they share a common concern with the temporal evolution of the interaction: how a body moves toward, and through, an interaction over time.

The temporal setting is orthogonal to the problem of this thesis, which synthesizes a single static interaction state rather than a motion sequence. A static state is not a degenerate motion; it is a different target, in which the body must be resolved into a specific configuration in metric contact with the scene, with no trajectory to average over or to absorb small errors. The dynamic methods above are therefore complementary to, rather than competitors of, the present work, and we do not pursue the temporal axis further. What is relevant is the observation that recurs across both the static and dynamic settings.

Control granularity: object and action, not surface region. In each of the methods above, semantic control resolves the interaction to the level of an object and an action. The text or label determines that the person sits on *this chair*, opens *this door*, or walks to *this table*, and language grounding, where present, is used precisely to identify which object in the scene the description refers to. What none of these methods produces is the finer quantity that plausibility ultimately depends on: the specific region of the identified object at which each body part makes contact. A chair affords sitting, but the seat region that the hips meet, and the armrests that the hands may rest on, are distinct surfaces that a pose-consistent interaction must resolve; a treadmill affords running, but the side rails the hands hold and the belt the foot lands on are different regions of the same object. Object-level semantic control selects the object; it does not localize contact on the object's surface, and it does not establish the per-body-part correspondence identified in section 2.1. Closing that gap requires reasoning below the object level, at the level of surface regions and the body parts that meet them, which is where foundation-model and zero-shot approaches, examined next, begin to operate.

2.3 Foundation-Model and Zero-Shot Interaction Synthesis

The methods of the previous sections learn interaction from paired data, which ties them to the scenes, objects, and interactions their training sets contain. A more recent line of work removes this dependence by treating large pretrained foundation models, image generators and vision-language models, as sources of interaction knowledge. Trained on vast quantities of images and text depicting people acting in the world, these models carry an implicit prior over how humans interact with their surroundings, and this prior can be distilled into 3D without any interaction-specific training. This is the zero-shot setting adopted in this thesis, and this section examines how existing methods use foundation models, what they leave unresolved, and why scene-side contact localization remains the open problem.

Grounding presupposes a contact region. Foundation models reason in their native spaces, an image generator produces pixels, a vision-language model produces text, and neither directly produces a body in metric contact with a reconstructed mesh. Bridging this gap is a problem of *grounding*: transferring the model's 2D or linguistic output into the 3D scene coordinate frame. The natural tool for grounding is the calibrated camera, which projects between the image and the scene, and this projection is indeed how image-space evidence is ultimately lifted to the mesh, including in the method of this thesis. But projection presupposes what it is meant to deliver: to project a contact onto the scene surface, one must already know *which pixels* correspond to that contact. A calibrated camera can map a 2D contact region to 3D, but it cannot decide, on its own, where in the image the contact occurs, which body part it belongs to, or which object surface it lies on. Identifying that region in the image is precisely scene-side contact localization, and it is the step that the projection depends upon but does not perform. The methods below can therefore be read according to how, and how reliably, they obtain the region that grounding requires.

Distilling interaction from vision-language models. GenZI [2] was the first zero-shot approach to 3D human–scene interaction synthesis. Given a scene, a text description, and a coarse location, it prompts a vision-language model to inpaint plausible humans into several rendered views of the scene, and then optimizes a 3D body so that its reprojection is consistent with the generated 2D people across views. The interaction knowledge comes entirely from the foundation model, and no 3D interaction data is used. GenZI thus establishes the core zero-shot recipe on which later work, including this thesis, builds: imagine the interaction in image space, then lift it to 3D. Its grounding, however, treats the generated image as an *implicit* whole. The 3D body is fit to the overall appearance of the inpainted humans, rather than to explicit, verified scene-side contact regions extracted from the image. There is no stage that isolates *this* pixel region as the surface the left hand contacts and *that* region as the surface the foot contacts, and no mechanism that checks whether the recovered contacts fall on the correct object surfaces. As a result, contact is only as reliable as the optimizer's implicit reading of an unstructured image, and errors, a hand grounded on the wrong surface, a contact that is plausible in the picture but wrong on the mesh, have no dedicated stage at which they can be detected and corrected.

Functional grounding by symbolic element identification. FunHSI [3] targets functionally correct interactions from open-vocabulary prompts, and is the closest prior work to this thesis in both goal and setting. Like the present method, it is training-free and reasons about human–scene contact through a graph structure; from a task prompt and a set of posed RGB-D views, it performs functionality-aware contact reasoning, identifies the functional scene elements involved, reconstructs their 3D geometry, and models the interaction through a contact graph, then synthesizes a body and refines it by optimization. For an instruction such as changing a room's temperature, for example, it identifies the thermostat as the functional element and grounds the interaction there. The shared use of a contact graph makes the true point of difference easy to misidentify: it is not the presence of a graph, but *how the scene side of the contact is resolved*. FunHSI grounds contact by identifying functional elements *symbolically and in advance*, deciding that the interaction involves a particular element and reconstructing that element's geometry as the contact target.

This strategy is well suited to small, localized functional elements, a thermostat, a knob, a handle, where identifying the element effectively determines where the contact falls, because the element and the contact region nearly coincide. It is weaker, however, precisely where the granularity argument of section 2.2 bites: identifying the element under-determines the contact location when the element is large. Knowing that the interaction involves a tabletop, a countertop, a treadmill rail, or a sofa seat does not by itself say *where along* that surface a hand should rest or a body should settle, yet it is that pose-consistent surface patch, not the element as a whole, that the contact must be grounded on. Reconstructing the entire element as the contact target leaves the precise region unresolved on exactly the surfaces where it matters most.

The proposed method deliberately resolves the scene side the other way. Its interaction graph names only the coarse contact structure, which body part contacts the target object, and withholds any commitment to a named functional part; the precise surface region is then recovered later, *visually*, from the generated interaction image and localized in the calibrated view, and it is accepted only after an explicit verification step. Because the region is localized directly in the image rather than inferred from the identity of a named element, the approach is indifferent to whether the functional surface is a small knob or a large rail: in both cases it localizes the specific patch the interaction requires. Where FunHSI resolves the scene side up front through symbolic element identification, the present method resolves it downstream through verified visual localization. The two approaches also differ in their inputs: FunHSI consumes multiple posed RGB-D views, whereas the present method operates from a single posed RGB view together with the reconstructed mesh.

Agentic generation and verification. The idea of not trusting a single foundation-model generation, but instead verifying it and correcting it in a loop, has recently emerged as a principle in its own right. WorldAgents [16] investigates whether 2D foundation image models can act as 3D world models, and finds that a single generation is unreliable; its answer is an agentic architecture in which a director formulates prompts, a generator synthesizes views by sequential inpainting, and a vision-language verifier evaluates the results in both 2D image space and 3D reconstruction space, curating and correcting the output. The lesson, that foundation-model output for 3D tasks becomes reliable when placed inside a generate-verify-correct loop rather than consumed in one shot, applies directly to contact localization. A single inpainted contact annotation may place a mask on the wrong surface, omit a required contact, or draw one of the wrong size; a closed-loop verifier that inspects the annotation against the intended interaction and issues corrections converts an unreliable single generation into a dependable one. This agentic verify-and-correct principle is central to the contact-estimation stage of this thesis, which uses exactly such a loop to obtain contact regions that a single generation could not be trusted to produce.

Zero-shot human–object interaction. A related group of methods applies foundation-model priors to human–*object* interaction (HOI), synthesizing a person interacting with a single object in isolation rather than within a scene. InteractAnything [17] uses large language models to reason about human–object relations and a 2D diffusion model to parse object affordances and extract contact points for open-set objects. HOI-PAGE [18] distills part affordance graphs from large language models, encoding how specific human body parts engage with specific object parts, and uses them to guide zero-shot 4D HOI synthesis. InterFusion [19] retrieves text-conditioned human poses and uses them as geometric priors for generating the interacting object. HOI-PAGE's part affordance graph is particularly close in spirit to the interaction graph used here, in that both encode part-level contact relations; the difference is that HOI-PAGE reasons over the parts of an isolated, given object, whereas the present method must localize contact on a surface embedded in a cluttered, reconstructed scene, viewed through a calibrated camera, where the target surface is neither isolated nor pre-segmented. This scene-embedded setting is what makes scene-side contact localization necessary: the contact region cannot be assumed given with the object, but must be found within the scene.

Summary. Across these methods, foundation models supply increasingly capable interaction priors, and the zero-shot recipe of imagining an interaction and lifting it to 3D is now well established. What remains unsolved is the reliable recovery of the scene-side contact region: GenZI leaves it implicit in the generated image, FunHSI fixes it symbolically before seeing the rendered interaction and under-determines it on large functional surfaces, and object-centric HOI methods sidestep it by assuming the object in isolation. None produces precise, verified, per-body-part contact regions on the surface of an object embedded in a reconstructed scene. The following section states this gap directly and positions the proposed method as its answer.

2.4 Positioning and Research Gap

The preceding sections trace a single thread. Contact is what makes an interaction plausible (section 2.1); semantic and language control determine which object and which action are involved but not where on the surface the contact falls (section 2.2); and foundation-model methods supply strong zero-shot interaction priors yet leave the scene-side contact region either implicit, symbolically pre-committed, or assumed given with an isolated object (section 2.3). Read together, these lines of work converge on one unmet capability.

The gap. No existing method produces *precise, verified, per-body-part contact regions on the surface of an object embedded in a reconstructed scene, in a zero-shot manner.* Each qualifier marks a distinct shortfall in prior work. *Precise* rules out object- and element-level grounding, which identifies the object or functional element but not the pose-consistent patch on its surface, and which under-determines the contact on large surfaces. *Verified* rules out single-shot generation, whose contact annotations may fall on the wrong surface, omit a required contact, or be wrongly sized, with no stage to detect and correct such errors. *Per-body-part* rules out body-side-only contact representations and whole-body placement, which do not pair each named part with its own scene-side region. *On a reconstructed scene* rules out object-centric HOI, which assumes the target in isolation rather than embedded in cluttered, calibrated scene geometry. And *zero-shot* rules out the data-driven contact predictors and scene-aware synthesis models that require captured paired data and do not transfer to unseen scenes and open-vocabulary instructions. It is the conjunction of these qualifiers, not any one alone, that remains unaddressed.

The proposed answer. This thesis addresses the gap through an *agentic contact estimation* procedure, which is the central contribution of the method. Rather than committing to a functional element symbolically or reading contact implicitly from a generated image, the method localizes each contact region directly in the calibrated view and accepts it only after explicit verification. An image generation model proposes body-part-specific contact regions on the scene; these regions are extracted and composited onto the original posed view, so that they remain in exact correspondence with the calibrated camera and the reconstructed mesh; and a vision-language evaluator inspects each composite against the intended interaction and, when necessary, issues a correction for another round. The result is a set of precise, per-body-part contact regions that have been checked for correctness before they are projected to 3D and used to drive optimization. Because the region is localized visually rather than derived from the identity of a named element, the procedure resolves contact on small and large functional surfaces alike; because it is verified in a closed loop, it converts an unreliable single generation into a dependable contact target; and because it relies only on pretrained foundation models and explicit camera geometry, it does so without any interaction-specific training. The symbolic interaction graph that seeds this process plays a deliberately minor role: it fixes only the coarse contact structure, which body part touches the target object, and leaves the precise surface region to be recovered visually.

Scope. The method draws a deliberate boundary between object contact and floor support. Precise scene-side localization through the agentic loop is applied to *target-object* contacts, where the specific surface patch is what makes the interaction correct, a hand on a particular rail, a foot on a belt. Floor support is handled separately, by segmenting the floor region directly, because a support contact requires a geometrically valid surface rather than a precise functional patch: the optimizer determines the exact foot placement on the floor, and no fine-grained localization of a functional region is needed. This division reflects where precision is informative and where it is not, and it is stated here so that the contribution is understood as precise contact localization on object surfaces, with floor support treated by simpler and sufficient means.

Positioning. Table 2.1 situates the proposed method against representative prior work along the qualifiers above. Data-driven contact and placement methods are not zero-shot; semantic-control methods operate at the object level; GenZI is zero-shot but leaves contact implicit; FunHSI grounds contact functionally but commits to elements symbolically and does not localize the patch on large surfaces; and object-centric HOI methods assume the object in isolation. The proposed method is the only entry that satisfies every qualifier simultaneously, which is the sense in which it occupies a previously empty position rather than improving incrementally on an existing one.

In summary, the proposed method is positioned as the first to deliver precise, verified, per-body-part scene-side contact localization for zero-shot static human–scene interaction synthesis on reconstructed scenes. The remainder of this thesis develops this method in detail and evaluates it against the prior work reviewed here.

Table 2.1: Positioning of the proposed method against representative prior work. Columns correspond to the qualifiers of the research gap: *Zero-shot* (no interaction-specific training); *Static scene* (targets a full 3D scene rather than an isolated object); *Scene-side contact* (localizes the contact region on the scene surface, not only on the body); *Precise patch* (resolves the specific surface region, not merely the object or element); and *Verified* (checks and corrects the contact before use). ✓ indicates the property holds, ✗ that it does not, and ▲ that it holds only partially, for example localizing contact to a functional element but not to the precise surface patch on large surfaces.

Method	Zero-shot	Scene not object	Scene-side contact	Precise patch	Verified
POSA [6]	✗	✓	✗	✗	✗
RICH / BSTRO [7]	✗	✓	✗	✗	✗
COINS [10]	✗	✓	✗	✗	✗
InteractAnything [17]	✓	✗	✓	▲	✗
HOI-PAGE [18]	✓	✗	✓	✓	✗
GenZI [2]	✓	✓	✗	✗	✗
FunHSI [3]	✓	✓	✓	▲	✗
Ours	✓	✓	✓	✓	✓

Chapter 3

Methodology

This chapter presents the proposed zero-shot pipeline for static human–scene interaction synthesis, which converts a natural-language instruction, a single posed ScanNet++ RGB view, and a reconstructed scene mesh into one SMPL-X body placed in the scene coordinate frame. Section 3.1 gives an overview of the pipeline and its five modules, and section 3.2 states the problem formulation and notation. The remaining sections describe the modules in turn: Scene Interaction Graph generation (section 3.3), human frame generation (section 3.4), agentic contact estimation (section 3.5), SMPL-X pose initialization (section 3.6), and scene-aware optimization (section 3.7).

3.1 Overview

This chapter presents a zero-shot pipeline for static 3D human–scene interaction synthesis. The method is zero-shot in the sense that it requires no interaction-specific training: it relies only on pretrained foundation models, together with explicit camera geometry and optimization. Its input is a natural-language instruction y, a single posed RGB view I of a ScanNet++ scene with known intrinsics K and extrinsics (R_c, t_c), and the reconstructed scene mesh M_s. Its output is a single SMPL-X body placed in the scene coordinate frame. The result is a *static interaction state*: one body configuration that realizes the requested interaction, rather than a motion sequence or an object-state change. For object-manipulation instructions, the object itself remains fixed in the reconstructed scene.

Throughout this chapter, a single running example illustrates each stage: the instruction "a person running on the treadmill, one foot landing on the moving belt, the other leg lifted behind, and both hands lightly holding the side handles." This example exercises the full method, since it requires two hand–object contacts, an asymmetric foot contact on a functional surface (the belt), and a lifted leg with no support.

Interaction instructions vary in specificity, and the method is designed to span this range.

An underspecified instruction such as “a person lifting a table” fixes the interaction only at a high level and leaves the body configuration and contact layout to be inferred. A functional instruction such as “a person opening a washing machine” additionally requires reasoning that contact should occur at a functional region, near the door or handle, rather than anywhere on the object. A detailed instruction, such as the treadmill example above, constrains the pose and the individual body-part contacts explicitly. In every case the underlying problem is the same: the instruction must be converted into metric 3D contact constraints that are consistent with the reconstructed scene.

A natural approach to solve this problem is to inpaint a human performing the interaction into the input view and then recover that person’s pose. This yields a body that looks correct in the image, but a pose recovered from a single view is expressed in image space and carries no guarantee of metric consistency with the scene: the recovered body may float above the floor, intersect furniture, or contact the wrong object region in 3D. The method’s key idea is therefore to use the inpainted frame not only to recover the body but also to *localize* fine-grained scene-side contact regions, which are grounded on the reconstructed mesh and used to drive SMPL-X optimization so that the final body respects the scene geometry.

Figure 3.1 shows the complete pipeline, which comprises five modules with the data flow

$$\text{SIG} \rightarrow \text{human frame} \rightarrow \left\{ \begin{matrix} \text{contact estimation} \\ \text{pose initialization} \end{matrix} \right\} \rightarrow \text{optimization.}$$

The first module converts the instruction and the view into a *Scene Interaction Graph* (SIG), which specifies the target object or floor, the relevant human body parts, and the required body-part–scene contact edges. The second module uses an image generation model to insert a single plausible human into the view according to the interaction instruction, producing a visual interaction hypothesis that captures pose, orientation, laterality, and approximate contact layout. This generated human frame then feeds two branches that run in parallel. In the first branch, the third module estimates scene-side contact regions: it marks each required contact on the original view with a distinct color to obtain body-part-specific contact masks, obtains floor-support regions separately with SAM3 when the interaction requires them, and projects the accepted masks onto the reconstructed mesh to form 3D contact targets. In the second branch, the fourth module initializes the body by passing the generated frame to GVHMR, which predicts an SMPL-X body conditioned on the camera focal length of the posed view. The fifth module refines this initialization through scene-aware optimization, holding the scene geometry and the projected contact targets fixed while optimizing the SMPL-X translation, orientation, pose, and a bounded global scale under contact and penetration data terms together with pose, orientation, height, and self-intersection priors.

Methodology pipeline placeholder (treadmill running example)

posed view I + instruction y $\rightarrow$ SIG $\rightarrow$ human frame $\hat{I}$ $\rightarrow$ contact masks $\rightarrow$ 3D contact targets P_e^s $\rightarrow$ GVHMR init θ_0 $\rightarrow$ optimized body θ^*

Figure 3.1: Overview of the proposed zero-shot static HSI synthesis pipeline. Given a natural-language instruction, a posed ScanNet++ RGB view, and the reconstructed scene mesh, the method generates a Scene Interaction Graph, synthesizes a human interaction image, estimates body-part-specific scene contact regions on the original view, projects these regions onto the scene mesh, initializes an SMPL-X body with GVHMR, and refines it through scene-aware optimization.

The full prompts used for the foundation-model stages are listed in appendix A.1, and the concrete model versions, hyperparameters, and iteration budgets are reported in the implementation details of **??**. This chapter describes the algorithmic role of each component; the appendix and the experiments chapter record the exact prompt wording and numerical settings.

3.2 Problem Formulation

Given a natural-language interaction instruction, a posed RGB view, and a reconstructed scene mesh, the goal is to synthesize a single human body that realizes the requested interaction in the scene coordinate frame. Let $I \in \mathbb{R}^{H\times W\times 3}$ denote the RGB view, which the user selects from the ScanNet++ scene. The view is calibrated: its intrinsics are given by the matrix K, and its pose in the scene coordinate frame is given by the rotation R_c and the translation t_c. The scene is represented by the triangle mesh $M_s = (V_s, F_s)$, with vertices $V_s \in \mathbb{R}^{N_s\times 3}$ expressed in the scene coordinate frame and triangular faces F_s.

The instruction y may range from underspecified to detailed, as in the examples of section 3.1. In all cases the method must infer a body configuration consistent with both the instruction and the scene geometry.

The output is a posed SMPL-X body expressed in the same coordinate frame as the reconstructed scene. SMPL-X is a differentiable parametric body model whose parameters are denoted

$$\theta = (\mathbf{T},\, \mathbf{r},\, \mathbf{p},\, \boldsymbol{\beta},\, s), \tag{3.1}$$

where $\mathbf{T}$ is the global translation, $\mathbf{r}$ the global orientation in axis-angle form, $\mathbf{p}$ the articulated body pose, $\boldsymbol{\beta}$ the body shape, and s a global isotropic scale applied to the body about its translation. Evaluating the model yields the human mesh $M_h(\theta) = (V_h(\theta), F_h)$ with posed vertices $V_h(\theta) \in \mathbb{R}^{N_h \times 3}$ and fixed faces F_h.

The synthesis target is a single static interaction state rather than a motion sequence. For object-manipulation instructions, the object remains in its reconstructed configuration; the method synthesizes the body that makes the interaction geometrically and visually plausible and does not simulate any subsequent object motion or state change.

The instruction is converted into a set of body–scene contact requirements, represented by a Scene Interaction Graph

$$G = (\mathcal{H},\, \mathcal{S},\, \mathcal{E}), \tag{3.2}$$

where $\mathcal{H}$ is the set of human body-part nodes, $\mathcal{S}$ the set of scene nodes, and $\mathcal{E}$ the set of contact edges. Human nodes are drawn from a fixed body-part vocabulary (hands, arms, feet, legs, hips, back, and head), and scene nodes are either the target object or the floor. Each edge

$$e = (h_e,\, s_e), \qquad e \in \mathcal{E}, \tag{3.3}$$

requires body part h_e to contact scene element s_e.

Each human node h_e maps to a fixed subset of SMPL-X contact vertices $B_e \subset \{1, \ldots, N_h\}$, defined once for the whole method. Under a body configuration θ, the corresponding body-side contact points are

$$P_e^h(\theta) = \{\, V_h(\theta)_i \mid i \in B_e \,\}. \tag{3.4}$$

The scene-side contact target for edge e is not annotated manually. It is estimated from generated visual contact annotations and projected onto the reconstructed mesh, yielding a set of scene points $P_e^s \subset \mathbb{R}^3$.

The body is initialized from the generated human frame using GVHMR, giving an initial estimate $\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \boldsymbol{\beta}_0, s_0)$. During optimization, the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, body pose $\mathbf{p}$, and global scale s are updated, while the shape $\boldsymbol{\beta}_0$ is taken from GVHMR and held fixed. Fixing $\boldsymbol{\beta}_0$ prevents the optimizer from distorting body identity to satisfy contact, while the single scale degree of freedom absorbs the residual overall size error inherent in monocular recovery; together, contact is explained through placement, pose, and a bounded change of overall size rather than through a change of body identity.

The synthesis problem is then to find SMPL-X parameters whose body-side contact points $P_e^h(\theta)$ approach the corresponding scene-side targets P_e^s, while the body remains physically plausible and avoids penetrating the scene. This is expressed as a scene-aware objective $\mathcal{L}_{\text{total}}$ that combines *data terms*, which tie the body to the observed scene through contact and penetration, with *regularization terms*, which keep the body plausible and close to the GVHMR initialization; both groups are defined in section 3.7. The final body is obtained by

$$\theta^* = \arg\min_{\mathbf{T},\mathbf{r},\mathbf{p},s} \mathcal{L}_{\text{total}}(\theta), \qquad \boldsymbol{\beta} = \boldsymbol{\beta}_0. \tag{3.5}$$

The following sections describe how the graph G, the human initialization θ_0, and the scene-side contact targets P_e^s are obtained.

3.3 Scene Interaction Graph Generation

The first module converts the open-language instruction into an explicit, symbolic specification of the required contacts. Its purpose is not to solve the geometry of the interaction but to determine which human body parts should contact which scene elements. This intermediate representation, the Scene Interaction Graph (SIG), provides the structure that later guides both contact-region estimation and SMPL-X optimization. The prompt used for SIG generation is given in appendix A.1; the description here concerns only the functional role of the module and the structure of its output.

The SIG is generated from the instruction y and the posed view I. A vision-language model is prompted to identify the target object, the human body parts that are physically relevant to the interaction, and any required floor support, and to return two outputs: the graph $G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$ of equation (3.2), and a detailed interaction description $\hat{y}$ that elaborates the original instruction. In the graph, each human node names a supported body part, each scene node names the target object or the floor, and each interaction edge pairs a human node with a scene node. The description $\hat{y}$ expands the terse instruction y into a fuller account of the interaction, specifying the pose and the individual per-body-part contacts (for each hand, foot, and so on) consistent with the graph edges, together with appearance attributes of the person such as clothing and hair. The graph fixes the symbolic contact structure used by contact estimation and optimization, while $\hat{y}$ conditions the image-generation stage that follows (section 3.4); the appearance attributes are included because they improve the realism and stability of the generated frame, on which the downstream pose recovery depends.

Body-part vocabulary. The human nodes are restricted to a fixed vocabulary: left and right hand, left and right arm, left and right leg, left and right foot, head, hips, and back. We fix this vocabulary so that every human node maps to a predefined SMPL-X contact region, a vertex subset $B_e \subset \{1, \ldots, N_h\}$ on the body mesh (figure 3.2). Hand nodes map to palm-side vertices, foot nodes to the sole-facing vertices, and sitting or leaning nodes to the corresponding regions on the hips, back, head, or legs. These subsets are defined once from the SMPL-X template and reused for all interactions. The restriction guarantees that

SMPL-X contact-region placeholder

Fixed contact-vertex subsets B_e on the SMPL-X template,
color-coded per body-part node in the SIG vocabulary
(treadmill example activates left hand, right hand, left foot)

Figure 3.2: Predefined SMPL-X contact regions for the supported body-part vocabulary. Each body-part node in the Scene Interaction Graph maps to a fixed vertex subset B_e on the SMPL-X template, shown here color-coded. These subsets are defined once and reused for all interactions; the treadmill running example activates the left-hand, right-hand, and left-foot regions.

every edge in a valid graph is grounded on one known body-side vertex set, which keeps the downstream optimization well defined and prevents the model from introducing body parts that cannot be attached to the body mesh.

Scene nodes and functional regions. Scene nodes are deliberately restricted to the target object and, when needed, the floor. The graph does not name object parts such as handle, knob, rung, or edge, and it does not attempt to localize them. An edge specifies only that a given body part contacts the target object; where on the object that contact should occur is left to the description $\hat{y}$ and to the later image-space estimation stage (section 3.5). This separation keeps the symbolic graph compact and robust: the language model is asked only for the coarse contact structure, which it can produce reliably, while the fine-grained localization of functional surfaces is deferred to a stage that has access to the rendered interaction and the calibrated view.

Table 3.2 shows the SIG produced for the treadmill running example. The instruction specifies both hands on the side handles and one foot on the belt, with the other leg lifted; the resulting graph contains two hand–treadmill edges and one foot–treadmill edge, and no floor edge, since the lifted leg provides no support and the supporting foot rests on the belt rather than the floor. The functional targets (handles, belt) appear nowhere in the graph itself; they are carried by the description $\hat{y}$ and localized later in image space. For the treadmill example, $\hat{y}$ reads, in part, “a person runs on the treadmill in a forward-moving pose, the left hand lightly holding the left side handle and the right hand the right side handle, the left foot landing on the belt while the right leg lifts behind,” together with appearance attributes such as tied-back hair, an athletic top, leggings, and running shoes.

This description drives the human-frame generation stage, while the graph in table 3.2 drives contact estimation and optimization. For contrast, a functional instruction such as "a person opening a washing machine" yields a single hand–washing-machine edge together with two foot–floor support edges, illustrating how the same structure spans instructions of differing specificity.

Validation. A generated graph is validated before it is used by later modules. The validation checks that the graph contains at least one scene node, that every human node belongs to the supported vocabulary, and that every edge references a defined scene node. Duplicate edges are merged, and edges with unsupported body parts are rejected. If no usable contact edge remains, the graph cannot yield meaningful contact targets and is regenerated. A validated graph therefore guarantees that each edge maps to exactly one body-side contact set B_e and one scene node, ready for the contact-estimation and optimization stages.

In summary, the SIG links language to geometry in a controlled way. It fixes which body parts participate and which scene element each of them contacts, and it produces an elaborated description $\hat{y}$ of the interaction, but it deliberately leaves two things unresolved: the exact object surface to be touched, and the final body pose. The next module uses $\hat{y}$ to generate a visual hypothesis of the interaction, from which the following modules localize the scene-side contact regions and initialize the body.

3.4 Human Frame Generation

The second module produces a visual hypothesis of the requested interaction in the input view. Given the posed view I and the detailed interaction description $\hat{y}$ from the SIG, an image generation model inpaints a single human into the scene so that the person appears to perform the interaction while the scene itself is left unchanged. The result is the *human frame*, an image in the same coordinate system as I that shows the hypothesized pose, orientation, laterality, and approximate contact layout. This frame is the shared input to the two parallel branches of the pipeline: it provides the image evidence from which GVHMR initializes the body (section 3.6), and it shows the interaction from which the scene-side contact regions are estimated (section 3.5). The prompts used in this module are given in appendix A.1.

Because both branches derive from the same human frame, the initialized body and the estimated contacts describe one and the same hypothesized interaction. This shared origin is what makes the two consistent when they are later combined: the contact targets and the initial pose are not independent estimates that must be reconciled, but two readings of the same generated evidence.

Preserving the scene. The generation prompt is constructed to keep the input view intact apart from the inserted person. The camera viewpoint, room geometry, object layout, lighting, textures, and target object are all to be preserved, and exactly one person is inserted. For object-manipulation instructions, the target object must not be moved, opened, lifted, rotated, or deformed. This constraint matters because the reconstructed mesh M_s represents the object in its original configuration and cannot reflect an image-only edit: for "a person opening a washing machine," the desired frame shows the person reaching toward the closed door or handle, not a washing machine whose door has been drawn open. For the treadmill example, the constraint keeps the treadmill in place and undeformed while the running person is inserted upon it.

Respecting the contact structure. The human frame must also be consistent with the contacts encoded in the SIG. Hand–object edges call for the corresponding hands to be placed near the target object, foot–floor edges call for the person to appear supported by the floor, and asymmetric contacts must be preserved with the correct laterality. For the treadmill example, the frame should show both hands on the side handles, the left foot on the belt, and the right leg lifted behind, matching the graph and the description $\hat{y}$. Preserving this structure is what makes the frame a useful bridge between the symbolic interaction and the metric optimization that follows.

Candidate selection. The image model produces a set of candidate frames

$$\mathcal{I}_H = \left\{ \hat{I}^{(1)}, \hat{I}^{(2)}, \dots, \hat{I}^{(N)} \right\}, \tag{3.6}$$

from which a vision-language evaluator selects the most suitable one. The evaluator compares each candidate against the input view I and the detailed instruction $\hat{y}$, and favors candidates that express the interaction clearly, preserve the target object and surrounding scene, exhibit a plausible full-body pose, and expose the relevant body-part contacts for later reasoning. Candidates with severe object distortion, missing body parts, implausible body scale, incorrect laterality, or unrelated interaction geometry are rejected. The selected frame is

$$\hat{I} = \hat{I}^{(j^\star)}. \tag{3.7}$$

Minor visual artifacts in $\hat{I}$ are tolerated, since the frame serves as evidence for pose and contact reasoning rather than as a final rendering; the selection step exists to ensure that both downstream branches operate on a frame that is visually consistent with the requested interaction. The number of candidates and the selection prompt are reported in **??** and appendix A.1, respectively.

The selected human frame $\hat{I}$ provides the visual evidence used by the following modules, while the authoritative scene geometry remains the reconstructed mesh M_s and the

Human frame generation placeholder (treadmill running example)

posed view I $\rightarrow$ candidate frames $\mathcal{I}_H$ $\rightarrow$ selected frame $\hat{I}$

Figure 3.3: Human frame generation and selection. The image model inpaints one person into the posed ScanNet++ view according to the interaction description $\hat{y}$, preserving the scene, camera viewpoint, and target object. A vision-language evaluator then selects the candidate that best preserves the scene and expresses the required body–scene contacts. The selected frame $\hat{I}$ is the shared input to GVHMR initialization and contact estimation.

authoritative scene appearance for contact annotation remains the original view I. The next section describes how the contact regions implied by $\hat{I}$ are localized on I and grounded on M_s.

3.5 Agentic Contact Estimation

The third module estimates the scene-side contact regions required by the interaction and grounds them on the reconstructed mesh. Given the human frame $\hat{I}$, the posed view I, and the graph G, it produces, for each contact edge, a verified 2D contact mask on the view I and projects it to the mesh to obtain the scene-side target P_e^s. This module is the bridge between the open-vocabulary visual reasoning of the foundation models and the metric optimization that follows: it is where the functional contact regions left unresolved by the SIG (the handles and the belt, in the treadmill example) are finally localized in the calibrated image and lifted into 3D.

The graph G re-enters the pipeline at this point. While human-frame generation is driven by the elaborated description $\hat{y}$, contact estimation is driven by the graph, since it needs the explicit per-edge structure: one mask, and later one 3D target, per contact edge.

Two distinct difficulties must be overcome, and the module addresses them by separate means. First, image generation is a powerful way to *reason* about where contact occurs but an unreliable way to *record* it: asked to paint contact regions onto the scene, an image model may silently alter the object, the background, or the image boundaries, and such drift would be fatal here, since the masks are projected to the mesh through the calibrated camera and must correspond pixel-for-pixel to the view I. This is handled structurally, by using the image model only to propose colored regions, extracting those colored pixels, and compositing them onto the untouched view I; the resulting *contact composite* carries the proposed contacts while retaining, by construction, the exact appearance and geometry of the calibrated view. Second, even on an untouched canvas the localization itself may be wrong: a required contact may be missing, a spurious one may appear, a mask may sit on the wrong object surface, or a footprint may be misplaced or wrongly sized. This is handled procedurally, by a closed-loop vision-language evaluator that inspects each composite and, when necessary, issues a correction for another round. The first mechanism guarantees geometric fidelity of the annotated image; the second, the *agentic* verify-and-correct loop, guarantees the correctness of the contacts drawn on it.

3.5.1 Contact-Overlay Generation

Contact estimation uses two images in complementary and authoritative roles. The human frame $\hat{I}$ is the *reference*: it is used solely to infer which body parts contact the target object and where those contacts fall on the object surface. The posed view I is the *canvas*: it is the base image and the authority for the target object's position, geometry, and boundaries. This division is enforced throughout the module: the reference determines *where contact occurs*, and the canvas determines *where the object is*, so that masks are always aligned to the geometry the projection step will use.

Each object-contact edge $e \in \mathcal{E}_{\mathrm{obj}}$ is assigned a distinct, saturated color c_e, where $\mathcal{E}_{\mathrm{obj}}$ is the subset of edges whose scene node is the target object. For the treadmill example, the left-hand, right-hand, and left-foot edges receive three different colors, so that their masks can be separated unambiguously after generation. At round r, the contact-overlay model receives the reference, the canvas, the graph, the color assignment, the previous composite $C^{(r-1)}$, and the correction instruction $a^{(r-1)}$, and produces an overlay candidate

$$O^{(r)} = \Phi_{\mathrm{con}}\big(\hat{I},\, I,\, G,\, \{c_e\},\, C^{(r-1)},\, a^{(r-1)}\big), \tag{3.8}$$

where Φ_{con} denotes the contact-overlay generation model. The first round uses neither a previous composite nor a correction instruction. When they are present, the model preserves the masks the evaluator has already accepted, copying them forward from $C^{(r-1)}$, and redraws only the regions the correction instruction identifies; this prevents already-correct masks from being regenerated and drifting between rounds.

Each mask is drawn as the localized *contact footprint* on the object surface, representing the apparent extent of contact rather than the silhouette of the touching body part, and every mask pixel is clipped strictly to the target-object boundary in the canvas, so that no mask extends onto the person, floor, wall, or background. The intended $O^{(r)}$ is thus a copy of the canvas bearing only these solid-color footprints; because it may nonetheless carry unintended edits to the scene, it is treated only as a source of colored regions, from which the masks are extracted next. The overlay prompt is given in appendix A.1.

3.5.2 Mask Extraction and Composite Construction

For each object-contact edge, the pixels matching the assigned color are extracted from the overlay candidate, yielding a binary mask

$$Z_e^{(r)} = \mathrm{ExtractColor}\big(O^{(r)},\, c_e\big). \tag{3.9}$$

Matching is performed within a small color tolerance, and small isolated components may be discarded. This step keeps only the painted footprints and discards everything else in $O^{(r)}$, including any incidental changes to object shape, texture, lighting, or background.

The extracted masks are then composited onto the untouched view I, producing the contact composite

$$C^{(r)} = \mathrm{Composite}\big(I,\, \{Z_e^{(r)}\}\big). \tag{3.10}$$

By construction, $C^{(r)}$ shares the resolution, camera calibration, and scene appearance of I, and differs from it only by the added contact overlays. This is what makes the composite, rather than the raw overlay, the authoritative annotated image: the image model is used only for its strength of proposing where contact occurs, while the calibrated view remains the sole geometric authority, and any scene drift in $O^{(r)}$ is discarded with the unmatched pixels.

3.5.3 Agentic Verification and Correction

The contact composite is verified by a vision-language evaluator in a closed loop. The evaluator receives the reference $\hat{I}$, the canvas I, and the current composite $C^{(r)}$: the reference is authoritative for which body parts contact the object and where, the canvas is authoritative for the object geometry, and the composite carries the masks under review. Crucially, a mask is not accepted merely because it lies plausibly on the object; it is accepted only if its location matches the contact shown in the reference, mapped onto the same object region in the canvas. Against this standard the evaluator checks for the characteristic failure modes: a missing required contact, an extra contact, a wrong body-part color, a mask on the wrong object surface, a mask shifted from the reference location, and a footprint of the wrong size. It returns a decision together with a natural-language correction instruction,

$$\big(d^{(r)},\, a^{(r)}\big) = \Phi_{\mathrm{eval}}\big(\hat{I},\, I,\, C^{(r)},\, G,\, \{c_e\}\big), \tag{3.11}$$

where $d^{(r)}$ is the accept/reject decision and $a^{(r)}$ is a single consolidated instruction covering every mask to be corrected; for example, moving a hand mask onto the correct handle, shrinking an oversized footprint, adding a missing contact, or removing a spurious one. Acceptance requires that all required contacts are present, no extra masks exist, every color is correct, and every footprint matches the surface, location, and size indicated by the reference. The evaluation prompt is given in appendix A.1.

On acceptance, the loop terminates and the current composite is used. On rejection, the correction instruction and the current composite are passed to the overlay model for the next round, giving the cycle

$$O^{(r)} \;\to\; \{Z_e^{(r)}\} \;\to\; C^{(r)} \;\to\; \big(d^{(r)}, a^{(r)}\big) \;\to\; O^{(r+1)}. \tag{3.12}$$

The loop runs for a fixed maximum number of rounds. It stops as soon as the evaluator accepts a composite; if the maximum is reached without acceptance, the masks from the final round are taken as the result, so that the procedure always returns a usable annotation. The maximum number of rounds is reported in **??**.

Let $r^\star$ denote the selected round. The final object-contact masks are

$$\Omega_e^{\mathrm{2D}} = Z_e^{(r^\star)}, \qquad e \in \mathcal{E}_{\mathrm{obj}}. \tag{3.13}$$

3.5.4 Floor-Support Masks

Floor contacts are handled separately from object contacts. For a support edge such as a foot on the floor, the precise pixel footprint matters less than a geometrically valid support surface, and asking the overlay model to paint a footprint on an empty stretch of floor invites hallucinated placement. Instead, SAM3 segments the visible floor in the view I, and the resulting region serves as the scene-side support surface; the optimizer then determines the exact foot placement on it, subject to the contact, penetration, and pose terms of section 3.7.

Let $\mathcal{E}_{\mathrm{floor}}$ denote the edges whose scene node is the floor, and let $\Omega^{\mathrm{2D}}_{\mathrm{floor}}$ be the segmented floor mask. Each floor-support edge takes this mask as its 2D contact region,

$$\Omega_e^{\mathrm{2D}} = \Omega^{\mathrm{2D}}_{\mathrm{floor}}, \qquad e \in \mathcal{E}_{\mathrm{floor}}, \tag{3.14}$$

and the same mask may be shared across several foot-support edges, since the distinction between left and right foot is enforced on the body side during optimization. The treadmill example has no floor edge, as its supporting foot rests on the belt; an interaction such as a person standing to open a washing machine, whose graph includes two foot–floor edges, would use this floor mask for both.

3.5.5 Projection to 3D Contact Targets

The verified 2D masks are lifted to the reconstructed mesh through the calibrated camera. A scene point $x \in \mathbb{R}^3$ in the scene coordinate frame projects to the image as

$$\tilde{u} = K\big(R_c x + t_c\big), \tag{3.15}$$

followed by perspective division to give the pixel coordinate u; write this calibrated projection as $\pi(x; K, R_c, t_c)$. A scene vertex or sampled surface point is assigned to edge e when its projection falls inside the mask Ω_e^{2D} and it is visible in the view. The scene-side contact target for edge e is therefore

$$P_e^s = \big\{\, x \in M_s \;\big|\; \pi(x; K, R_c, t_c) \in \Omega_e^{\mathrm{2D}} \text{ and } x \text{ visible} \,\big\}, \tag{3.16}$$

represented in practice as a set of scene vertices or sampled surface points. These points are fixed for the remainder of the pipeline and act as the regions that the corresponding SMPL-X contact vertices are drawn toward. Collecting the targets over all edges gives the output of the module,

$$\mathcal{P}^s = \big\{\, P_e^s \;\big|\; e \in \mathcal{E} \,\big\}, \tag{3.17}$$

in which each element is linked, through its edge, to one body-side contact set B_e. The next module initializes the SMPL-X body, after which optimization brings the two sides into contact.

3.6 SMPL-X Pose Initialization

The fourth module initializes the SMPL-X body from the human frame. This is the second of the two branches fed by $\hat{I}$: while contact estimation localizes the scene-side targets, this module recovers a complete articulated body to place against them. The scene-side targets constrain only where a few body regions should end up and do not by themselves determine a full pose. A good initial body is therefore essential, and not merely convenient: the scene-aware objective of section 3.7 is non-convex, so the quality of its solution depends heavily on where it starts. An initialization close to the intended interaction lets the optimizer make small, local adjustments, whereas a naive or arbitrary starting body would leave it prone to poor local minima. The method obtains this initialization with GVHMR.

GVHMR performs monocular human recovery, predicting an SMPL-X body in the camera frame. As it is trained for this task, the body it recovers is a realistic, well-articulated pose that already captures the coarse configuration of the interaction shown in the human frame: the bend of the limbs, the global orientation, and the overall body arrangement are close to correct before any scene-aware refinement. This is precisely the property the optimization needs. Because the initial body already sits near a plausible solution, the

Agentic contact estimation placeholder (treadmill running example)

reference $\hat{I}$ + canvas I $\rightarrow$ overlay $O^{(r)}$ $\rightarrow$ extracted masks $Z_e^{(r)}$ $\rightarrow$ composite $C^{(r)}$

evaluator: accept or correct $\rightarrow$ final masks Ω_e^{2D} $\rightarrow$ 3D targets P_e^s

Figure 3.4: Agentic contact estimation. Using the human frame $\hat{I}$ as reference and the posed view I as canvas, the overlay model paints colored contact footprints, of which only the colored pixels are extracted and composited back onto the untouched view I. A vision-language evaluator checks the composite against the reference and the canvas and either accepts it or returns a correction for another round. Accepted masks are projected through the calibrated camera to the reconstructed mesh to yield body-part-specific 3D contact targets P_e^s. For the treadmill example, the left hand, right hand, and left foot receive distinct colors and are localized on the two handles and the belt.

subsequent refinement adjusts placement and pose only locally, which keeps it clear of the poor local minima that a weaker initialization would invite.

Two calibration details keep this initialization metrically consistent with the rest of the pipeline. First, GVHMR is conditioned on the camera focal length, which is read from the intrinsics K of the posed view; supplying the true focal length rather than a default keeps the recovered body consistent with the calibrated camera. Second, the camera-frame estimate is transformed into the scene coordinate frame using the extrinsics (R_c, t_c) of the same view, so that the initial body is placed in the same frame as the reconstructed mesh and the projected contact targets, expressed through the very geometry used to project those targets.

Since the setting provides a single static frame rather than a video, the frame is repeated to form a short static clip,

$$\mathcal{V}_{\hat{I}} = \big(\hat{I},\, \hat{I},\, \ldots,\, \hat{I}\big), \tag{3.18}$$

which GVHMR processes as an ordinary sequence; the first recovered frame is taken as the initialization. Repeating the frame preserves the static nature of the target interaction while satisfying the model's sequence input.

Applying GVHMR to the clip yields an initial SMPL-X estimate in the camera frame,

$$\theta_0^{\mathrm{cam}} = \big(\mathbf{T}_0^{\mathrm{cam}},\, \mathbf{r}_0^{\mathrm{cam}},\, \mathbf{p}_0,\, \boldsymbol{\beta}_0\big), \tag{3.19}$$

with global translation $\mathbf{T}_0^{\mathrm{cam}}$ and orientation $\mathbf{r}_0^{\mathrm{cam}}$ expressed relative to the camera, articulated pose $\mathbf{p}_0$, and shape $\boldsymbol{\beta}_0$. The pose and shape are frame-relative rather than camera-relative and are carried over unchanged; only the global translation and orientation must be moved into the scene coordinate frame.

The optimization runs in the scene frame, so the initialization is transformed from camera to world coordinates using the pose of the selected view. With the convention

$$x^{\mathrm{cam}} = R_c\, x^{\mathrm{world}} + t_c, \tag{3.20}$$

a camera-frame point maps to the world frame by the inverse transform. The global translation is converted as

$$\mathbf{T}_0 = R_c^{\top}\big(\mathbf{T}_0^{\mathrm{cam}} - t_c\big), \tag{3.21}$$

and the global orientation by applying the inverse camera rotation. Writing $\mathbf{R}(\cdot)$ for the rotation matrix of an axis-angle vector, the world-frame orientation matrix is

$$\mathbf{R}(\mathbf{r}_0) = R_c^{\top}\,\mathbf{R}\big(\mathbf{r}_0^{\mathrm{cam}}\big), \tag{3.22}$$

whose axis-angle form is $\mathbf{r}_0$. The initialization in the scene frame is then

$$\theta_0 = \big(\mathbf{T}_0,\, \mathbf{r}_0,\, \mathbf{p}_0,\, \boldsymbol{\beta}_0,\, s_0\big), \tag{3.23}$$

where the global scale is initialized to $s_0 = 1$.

While the recovered pose is realistic, it is estimated from the human frame alone and is therefore unaware of the reconstructed scene: it is not guaranteed to satisfy the projected contact targets or to avoid intersecting the mesh, and the frame may carry small errors in scale, depth, or contact placement. The role of the final module is to refine the translation $\mathbf{T}_0$, orientation $\mathbf{r}_0$, pose $\mathbf{p}_0$, and global scale s_0 against the scene geometry and the contact targets, starting from this strong initialization. Throughout that refinement the shape $\boldsymbol{\beta}_0$ is held fixed, so that identity is preserved: contact is explained by placement, pose, and a bounded overall scale rather than by altering body identity, while the single scale degree of freedom absorbs the residual size error of the monocular estimate. The reconstructed mesh and the projected targets likewise remain fixed.

3.7 Scene-Aware SMPL-X Optimization

The fifth module refines the GVHMR initialization into the final body. The initialization is a plausible, well-articulated pose, but it is unaware of the scene: it need not satisfy the projected contact targets, and it may intersect the reconstructed mesh. This module corrects those errors by minimizing a scene-aware objective while holding the scene fixed.

The objective combines two *data terms*, which tie the body to the observed scene through contact and non-penetration, with four *regularization terms*, which keep the body plausible and anchored to the trusted initialization. The scene mesh M_s and the projected contact targets $\{P_e^s\}$ remain fixed throughout; only the body is optimized.

3.7.1 Optimized Variables

The optimization starts from the world-frame initialization $\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \boldsymbol{\beta}_0, s_0)$ of section 3.6 and updates the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, articulated pose $\mathbf{p}$, and global scale s, while the shape $\boldsymbol{\beta}_0$ is held fixed:

$$\theta = (\mathbf{T},\, \mathbf{r},\, \mathbf{p},\, \boldsymbol{\beta}_0,\, s). \tag{3.24}$$

Fixing $\boldsymbol{\beta}_0$ preserves body identity, so that contact is achieved through placement, articulation, and a bounded change of overall size rather than by distorting the body's proportions. The global orientation is optimized through a continuous six-dimensional rotation representation, which avoids the discontinuities of the axis-angle parameterization and yields a stable gradient for the orientation update. For any parameter state, the SMPL-X model produces the posed mesh $M_h(\theta) = (V_h(\theta), F_h)$, whose contact vertices are compared against the fixed scene targets.

3.7.2 Data Terms

Contact objective. The contact term attracts each body-side contact region toward its projected scene-side target. For edge e, the body-side contact points $P_e^h(\theta) = \{V_h(\theta)_i \mid i \in B_e\}$ are pulled toward the target P_e^s by a mean one-sided nearest-neighbor distance:

$$\mathcal{L}_{\text{contact}}(\theta) = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{1}{|B_e|} \sum_{i \in B_e} \min_{q \in P_e^s} \big\| V_h(\theta)_i - q \big\|_2^2. \tag{3.25}$$

The distance is measured in one direction only, from the body to the scene: each body contact vertex is drawn to its nearest point in the scene target, but scene points are under no obligation to be reached. This direction reflects the geometry of the problem. The body-side set B_e is a small, fixed region whose every vertex should make contact, whereas the scene-side target P_e^s is a broad projected region of which only a part is actually touched, since a palm covers a small patch of a handle and a sole a small patch of the belt. A symmetric distance, which would also pull each scene point toward the body, would therefore penalize the uncovered remainder of the target and try to spread the body across the entire region rather than let it rest naturally on a portion of it. Matching only from the body to the scene also removes any need for a fixed body-to-scene vertex correspondence and lets the target act as a region the contact vertices may settle anywhere within. Averaging over the whole set B_e, rather than taking a single closest vertex, then encourages the entire contact region to meet the surface: for the treadmill example, this draws the full palms toward the two handles and the sole toward the belt.

Penetration objective. The contact term alone can pull the body through the scene surface, so a penetration term discourages the body from occupying scene geometry. Using VolumetricSMPL [20], which equips the posed SMPL-X body with a differentiable signed-distance field, the method queries the signed distance $\phi_\theta(x)$ from each of a set of scene points $\mathcal{Q}_s$, sampled from the reconstructed mesh near the body, to the body surface, with $\phi_\theta(x) < 0$ indicating that x lies inside the body. The term penalizes scene points that fall inside the body or within a small clearance margin $\delta \geq 0$:

$$\mathcal{L}_{\text{pen}}(\theta) = \frac{1}{|\mathcal{Q}_s|} \sum_{x \in \mathcal{Q}_s} \operatorname{ReLU}\big(\delta - \phi_\theta(x)\big). \tag{3.26}$$

Points comfortably outside the body contribute nothing, so the term acts only where the body would otherwise intersect the scene, allowing contact while forbidding interpenetration.

3.7.3 Regularization Terms

Pose prior. Because the GVHMR pose is already realistic, the articulated pose is kept close to its initial value, so that contact is satisfied by small adjustments rather than by drifting into an unnatural configuration:

$$\mathcal{L}_{\text{pose}}(\theta) = \|\mathbf{p} - \mathbf{p}_0\|_2^2. \tag{3.27}$$

Orientation prior. The global orientation is likewise anchored to the initialization, penalized by the geodesic distance between the current and initial rotations:

$$\mathcal{L}_{\text{orient}}(\theta) = \big\|\operatorname{Log}\big(\mathbf{R}(\mathbf{r}_0)^\top\, \mathbf{R}(\mathbf{r})\big)\big\|_2^2, \tag{3.28}$$

where $\mathbf{R}(\cdot)$ is the rotation matrix of an axis-angle vector and $\operatorname{Log} : SO(3) \to \mathbb{R}^3$ returns the axis-angle vector of a rotation. This prevents the optimizer from satisfying contact through large, unrealistic global rotations inconsistent with the human frame.

Height prior. The global scale s is free to correct the overall size of the monocular estimate, but it must not drive the body to an implausible stature. Writing the body's metric height as $h(\theta) = s\, h_{\text{can}}$, where h_{can} is the height of the fixed-shape body at unit scale, the height prior keeps this stature near a target h_{tgt}:

$$\mathcal{L}_{\text{height}}(\theta) = \left(\frac{h(\theta) - h_{\text{tgt}}}{\sigma_h}\right)^2, \tag{3.29}$$

with σ_h setting the tolerated deviation. Together with a bounded parameterization of the scale (**??**), this term confines s to a realistic human height range, so that the size correction resolves the residual scale error of monocular recovery without introducing an implausibly small or large body.

Self-intersection prior. Finally, a self-intersection term discourages the body mesh from penetrating itself, for instance an arm passing through the torso. Following [21], the colliding triangles of the posed body are first detected with a bounding-volume hierarchy [22], giving a set $\mathcal{C}(\theta)$ of intersecting triangle pairs; a local conic distance-field penalty ψ then measures the interpenetration of each colliding pair and pushes them apart:

$$\mathcal{L}_{\text{self}}(\theta) = \frac{1}{|\mathcal{C}(\theta)|} \sum_{(a,b)\in\mathcal{C}(\theta)} \psi\big(T_a(\theta),\, T_b(\theta)\big), \tag{3.30}$$

where $T_a(\theta)$ and $T_b(\theta)$ are the posed triangles of a colliding pair and ψ vanishes when they are disjoint. Only colliding pairs contribute, so the term is active solely where the body would otherwise intersect itself, keeping the refined body physically coherent without affecting collision-free regions. The parameters of ψ are given in **??**.

3.7.4 Total Objective

The full objective is the weighted sum of the two data terms and the four regularization terms:

$$\mathcal{L}_{\text{total}}(\theta) = \underbrace{\lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{pen}}\mathcal{L}_{\text{pen}}}_{\text{data terms}} + \underbrace{\lambda_{\text{pose}}\mathcal{L}_{\text{pose}} + \lambda_{\text{orient}}\mathcal{L}_{\text{orient}} + \lambda_{\text{height}}\mathcal{L}_{\text{height}} + \lambda_{\text{self}}\mathcal{L}_{\text{self}}}_{\text{regularization terms}}, \tag{3.31}$$

and the final body is obtained by

$$\theta^* = \underset{\mathbf{T},\,\mathbf{r},\,\mathbf{p},\,s}{\arg\min}\, \mathcal{L}_{\text{total}}(\theta), \qquad \boldsymbol{\beta} = \boldsymbol{\beta}_0. \tag{3.32}$$

The data terms enforce the functional body–scene relationships inferred by the earlier modules while forbidding interpenetration, and the regularization terms keep the result close to the trusted initialization and within a plausible human range. The loss weights $\lambda_{(\cdot)}$ are reported in **??**.

3.7.5 Optimization Procedure

The objective is minimized with Adam in two stages, following a coarse-to-fine schedule that exploits the quality of the initialization. In the first, *rigid* stage, only the global translation and scale are optimized, while the orientation and articulated pose are frozen; this places and sizes the trusted GVHMR body against the scene without deforming its pose, and the self-intersection term is disabled since the pose cannot change. In the second, *pose* stage, all variables — translation, orientation, pose, and scale — are optimized under the full objective, allowing local articulation to refine the contacts. Separating placement from articulation keeps the optimizer near the plausible initialization for as long as possible and reduces the risk of settling in a poor local minimum.

At each iteration the SMPL-X mesh is recomputed, the body contact vertices are matched to their fixed scene targets, penetration is evaluated against the sampled scene points, and gradients are backpropagated through the differentiable body model; the scene mesh, the projected contact targets, and the body shape stay fixed. The optimizer returns the final parameters

$$\theta^* = (\mathbf{T}^*, \mathbf{r}^*, \mathbf{p}^*, \boldsymbol{\beta}_0, s^*), \tag{3.33}$$

and the final synthesized interaction is the posed mesh

$$M_h^* = M_h(\theta^*), \tag{3.34}$$

already expressed in the ScanNet++ scene coordinate frame. The stage lengths, learning rate, gradient clipping, and total iteration budget are reported in **??**.

Table 3.1: Notation used throughout the methodology chapter.

Symbol	Meaning
Inputs and scene	
I	Posed RGB view selected by the user from a ScanNet++ scene.
K	Camera intrinsic matrix.
R_c, t_c	Camera rotation and translation (extrinsics) in the scene frame.
$\pi(\cdot; K, R_c, t_c)$	Calibrated projection from the scene frame to the image plane.
$M_s = (V_s, F_s)$	Reconstructed scene mesh in the scene coordinate frame.
y	Natural-language interaction instruction.
$\hat{y}$	Detailed interaction description generated with the SIG.
Scene Interaction Graph	
$G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$	Scene Interaction Graph.
$e = (h_e, s_e)$	Contact edge between body part h_e and scene element s_e.
$\mathcal{E}_{\text{obj}}$, $\mathcal{E}_{\text{floor}}$	Object-contact and floor-support edge subsets.
Body and contact	
$\theta = (\mathbf{T}, \mathbf{r}, \mathbf{p}, \boldsymbol{\beta}, s)$	SMPL-X parameters: translation, orientation, pose, shape, scale.
$M_h(\theta) = (V_h(\theta), F_h)$	Posed SMPL-X mesh.
$\mathbf{R}(\cdot)$	Rotation matrix of an axis-angle vector.
$\mathrm{Log}(\cdot)$	Map $SO(3) \to \mathbb{R}^3$ returning the axis-angle vector of a rotation.
B_e	Fixed SMPL-X contact-vertex subset for the body part in edge e.
$P_e^h(\theta)$	Body-side contact points for edge e.
P_e^s	Scene-side contact target for edge e, on the reconstructed mesh.
Ω_e^{2D}	Verified 2D contact mask for edge e in the view I.
Foundation-model operators and intermediates	
$\hat{I}$	Selected generated human frame.
$\mathcal{I}_H$	Set of candidate human frames.
Φ_{con}, Φ_{eval}	Contact-overlay generation and contact-evaluation models.
c_e	Distinct color assigned to object-contact edge e.
$O^{(r)}$, $Z_e^{(r)}$, $C^{(r)}$	Overlay candidate, extracted mask, and composite at round r.
$d^{(r)}$, $a^{(r)}$	Accept/reject decision and correction instruction at round r.
Initialization	
θ_0^{cam}	GVHMR estimate in the camera frame.
θ_0	GVHMR initialization transformed to the scene frame.
s_0	Initial global scale, $s_0 = 1$.
Optimization	
$\phi_\theta(x)$	Signed distance from scene point x to the posed body.
$\mathcal{Q}_s$	Scene points sampled near the body for penetration evaluation.
δ	Clearance margin in the penetration term.
$h(\theta)$, h_{can}, h_{tgt}	Current, unit-scale, and target body heights.
σ_h	Tolerated height deviation in the height prior.
$\mathcal{C}(\theta)$, ψ	Colliding triangle pairs and their conic penetration penalty.

Table 3.2: Scene Interaction Graph for the treadmill running example. The graph records only the coarse body-part–object contacts; the functional target region (handle, belt) is carried by the description $\hat{y}$ and localized later in image space, rather than encoded as a graph node.

Human node	Scene node
left hand	treadmill
right hand	treadmill
left foot	treadmill

Chapter 4

Experiments

This chapter defines the experimental protocol used to validate the thesis claims.

4.1 Research Questions

State the questions the experiments answer. Each question should map to a table, figure, ablation, or qualitative analysis.

4.2 Datasets

Describe all datasets, splits, annotations, preprocessing steps, and any generated data.

4.3 Baselines

List baseline methods and explain how they were configured. Make comparisons fair and reproducible.

4.4 Metrics

Define quantitative metrics and explain what each metric captures. Include limitations of each metric where relevant.

Table 4.1: Example experiment configuration table. Replace with the final setup.

Item	Value
Dataset split	train / validation / test
Batch size	–
Optimizer	–
Hardware	–

4.5 Experimental Setup

Specify training schedule, optimizer, learning rate, batch size, number of runs, random seeds, hardware, and software versions.

Chapter 5

Results and Discussion

This chapter reports the main results, ablations, qualitative examples, and limitations.

5.1 Main Results

Present the primary quantitative results first. Explain the trend, not just the numbers.

5.2 Ablation Studies

Isolate the effect of each contribution. Keep ablations tied to the claims made in the introduction.

5.3 Qualitative Results

Show representative successes and failures. Qualitative figures should make the method's behavior inspectable.

5.4 Limitations

Discuss failure cases, data bias, compute cost, assumptions, and evaluation gaps. Strong limitations make the thesis more credible.

Table 5.1: Main result table placeholder.

Method	Metric 1	Metric 2	Metric 3
Baseline	–	–	–
Proposed	–	–	–

5.5 Discussion

Connect the results back to the motivation. Explain what the experiments imply for the broader research problem.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

Summarize the thesis in a few paragraphs: problem, approach, evidence, and main takeaway. Avoid introducing new experiments here.

6.2 Future Work

Describe concrete extensions that follow from the limitations and results. Separate near-term engineering improvements from longer-term research directions.

6.3 Closing Remarks

End with a brief statement about the expected impact or usefulness of the work.

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Appendix A

Additional Material

A.1 Foundation Model Prompts

This appendix records the system prompts used by the foundation-model stages in chapter 3. The symbols below are used in the methodology chapter.

Table A.1: System-prompt symbols used in the methodology.

Symbol	Stage	Source or status
Π_{SIG}	SIG generation	`4DHSI/01_Generate_SIG/system_prompt_sig.md`
Π_{H}	Human-frame inpainting	`4DHSI/02_Generate_Human_Frame/system_prompt_human_inpaint.md`
Π_{sel}	Best human-frame selection	Placeholder; final system prompt not fixed yet.
Π_{con}	Contact overlay generation	`4DHSI/03_Estimate_Contact_Agentic/system_prompt_estimate_contact_agentic.md`
Π_{eval}	Contact VLM evaluation	`4DHSI/03_Estimate_Contact_Agentic/vlm_prompt_evaluate_contact.md`

A.1.1 SIG Generation System Prompt Π_{SIG}

```
You generate a Scene Interaction Graph (SIG) for a static human-scene interaction.

Produce a compact graph containing the target object or objects, relevant human
body parts, direct physical contact edges, and a static interaction description.

Input:
- A scene image from the selected camera view.
- A JSON request:
```

```
{
  "interaction": "short interaction description"
}

Output only valid JSON. Do not include markdown or commentary.

Required output schema:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "short primary target object name"
    },
    {
      "id": "target_object_2",
      "label": "short secondary target object name"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "brief reason for this contact"
    }
  ],
  "interaction": "Static interaction description under 150 words."
}

Rules:

1. Use the interaction text to identify the intended human-scene contact, and use the
↪ image only to understand visible object context.
2. `target_objects` should contain the scene object that receive direct human contact
↪ in the static pose.
3. Use one target object when the contact is with a single object or multiple
↪ structural parts of the same object, such as a chair back, bed frame, or bathtub
↪ side.
4. Use two target objects only when two separate scene objects are directly contacted.
↪ For example, use "bathtub" and "curtain rail" when one foot contacts the bathtub
↪ and one hand contacts the curtain rail.
5. Keep target object labels short noun phrases.
6. Use human part nodes only from this vocabulary:
   - left hand
   - right hand
   - left arm
   - right arm
   - left leg
   - right leg
   - left foot
   - right foot
   - head
   - hips
   - back
```

```
7. Include a human part node only when that part is physically important for contact
↪  or pose.
8. `scene_nodes` can include only:
   - target_object_1
   - target_object_2
   - floor
9. Include `target_object_2` only if it exists in `target_objects` and has at least
↪  one direct contact edge.
10. `interaction_edges` represent direct physical contact only.
11. Use `scene_element: "target_object_1"` or `scene_element: "target_object_2"` for
↪  contacts with the corresponding target object.
12. Use floor edges only for required direct floor support. If there is no floor
↪  contact, simply omit `floor` from `scene_nodes` and `interaction_edges`.
13. Do not create object-part nodes.
14. Keep interaction edges focused on physically necessary contacts.
15. The output `interaction` must describe the static pose and physical contacts under
↪  150 words. It must explicitly restate every `interaction_edges` contact using the
↪  same side-specific human part names, such as "left hand", "right foot", or "back",
↪  and the contacted scene object label or floor. Avoid vague phrases such as "one
↪  hand", "both feet", or "legs" when the edge uses specific left/right parts. It
↪  should also describe clothing, shoes, hair, and facial expression.

Few-shot examples:

These few-shot examples are written without a scene image. They demonstrate the
expected JSON structure and explicit contact wording. For the real request, use
the provided scene image to decide the exact contact configuration, including
which side-specific hand, foot, leg, or arm is in contact.

Example 1 input JSON:

{
  "interaction": "a person lifting the brown small wooden table with both hands"
}

Example 1 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "brown small wooden table"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the table while preparing to lift"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
```

```
      "notes": "the right hand grips the opposite side of the table"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the standing pose on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the standing pose on the floor"
    }
  ],
  "interaction": "A person stands beside the brown small wooden table in a
  ↪  forward-leaning pose. The left hand grips one side of the brown small wooden
  ↪  table, the right hand grips the opposite side, the left foot is planted on the
  ↪  floor, and the right foot is planted on the floor. The person has short black
  ↪  hair, a neutral focused facial expression, a gray shirt, blue jeans, and white
  ↪  sneakers."
}

Example 2 input JSON:

{
  "interaction": "a person lying on bed with his head on the pillow"
}

Example 2 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bed with pillows"
    }
  ],
  "human_part_nodes": [
    "person 1, left leg",
    "person 1, right leg",
    "person 1, head",
    "person 1, hips",
    "person 1, back"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "head",
      "scene_element": "target_object_1",
      "notes": "the head rests on the pillow as part of the bed setup"
    },
    {
      "human_part": "hips",
      "scene_element": "target_object_1",
      "notes": "the hips are supported by the mattress"
    },
    {
      "human_part": "back",
      "scene_element": "target_object_1",
      "notes": "the back rests against the mattress"
    },
    {
```

```
      "human_part": "left leg",
      "scene_element": "target_object_1",
      "notes": "the left leg rests on the bed"
    },
    {
      "human_part": "right leg",
      "scene_element": "target_object_1",
      "notes": "the right leg rests on the bed"
    }
  ],
  "interaction": "A person lies on the bed with pillows in a relaxed static pose. The
  ↪  head rests on a pillow, the back rests on the mattress, the hips are supported
  ↪  by the mattress, the left leg rests on the bed, and the right leg rests on the
  ↪  bed. The person has short black hair, a calm sleepy facial expression, a gray
  ↪  shirt, soft pants, and socks."
}

Example 3 input JSON:

{
  "interaction": "a person stepping over the bathtub wall, one foot inside the tub,
  ↪  the other foot still on the floor, one hand holding the curtain rail for balance"
}

Example 3 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bathtub"
    },
    {
      "id": "target_object_2",
      "label": "curtain rail"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_2",
      "notes": "the left hand grips the curtain rail for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "target_object_1",
      "notes": "the left foot presses inside the bathtub while stepping over the wall"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot stays planted on the floor outside the bathtub"
    }
```

```
  ],
  "interaction": "A person steps over the bathtub wall in a careful balancing pose.
  ↪  The left hand grips the curtain rail, the left foot presses inside the bathtub,
  ↪  and the right foot is planted on the floor outside the tub. The person has short
  ↪  black hair, a focused facial expression, a gray shirt, rolled-up pants, and bare
  ↪  feet."
}

Example 4 input JSON:

{
  "interaction": "a person bending down in front of the lower drawer, one hand pulling
  ↪  the cabinet door open and the other hand holding the counter edge for balance"
}

Example 4 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "lower cabinet door"
    },
    {
      "id": "target_object_2",
      "label": "counter edge"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot",
    "person 1, hips"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand pulls the lower cabinet door"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_2",
      "notes": "the right hand holds the counter edge for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the squat on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the squat on the floor"
    }
  ],
```

```
  "interaction": "A person bends down in front of the lower cabinet door in a balanced
  ↪  pose. The left hand pulls the lower cabinet door, the right hand holds the
  ↪  counter edge, the left foot is planted on the floor, and the right foot is
  ↪  planted on the floor. The person has short black hair, a neutral concentrated
  ↪  facial expression, a gray shirt, blue jeans, and white sneakers."
}

Example 5 input JSON:

{
  "interaction": "a person hanging from the pull-up bar with both hands, elbows bent,
  ↪  knees tucked upward, and both feet off the floor"
}

Example 5 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "pull-up bar"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the pull-up bar"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
      "notes": "the right hand grips the pull-up bar"
    }
  ],
  "interaction": "A person hangs from the pull-up bar with elbows bent and knees
  ↪  tucked upward. The left hand grips the pull-up bar, the right hand grips the
  ↪  pull-up bar, the left foot stays off the floor, and the right foot stays off the
  ↪  floor. The person has short black hair, a strained focused facial expression, a
  ↪  gray athletic shirt, dark shorts, and sneakers."
}
```

A.1.2 Human-Frame Inpainting System Prompt Π_{H}

```
Input Image:

- The image is the original scene. Use the interaction description to identify the
↪  existing
target object that must remain fixed and contacted.

Scene Preservation Instructions:

- Use the provided scene image as a fixed camera reference.
```

```
- Insert exactly one person performing the described interaction with the target
↪  object.
- This frame must represent the pre-motion start state: the person is already in
↪  natural contact
with the target object of the scene, but no visible object motion has begun yet.
- Target object is already present in the scene. Do not add it.
- Preserve the camera pose, field of view, perspective, room geometry, lighting,
↪  shadows, textures,
and all existing object identities.
- Do not add new furniture or remove existing items.
- Keep the target object in exactly the same pose, position, orientation, scale,
↪  material,
and appearance as in the original scene.
- Do not move, rotate, deform, lift, translate, or reposition the target object in any
↪  way.
- Only add the human in plausible contact, as if about to start the action in the next
↪  frame.

Output:

- Return only the edited scene image.
```

A.1.3 Best Human-Frame Selection System Prompt Π_{sel}

```
PLACEHOLDER: Final best-frame selection system prompt not fixed yet.

Intended role:
- Given the original scene image, the interaction description or SIG description, and
↪  multiple human-inpainted candidate frames, select the single candidate used as the
↪  visual prior for GVHMR initialization and contact reasoning.

Intended ranking criteria:
1. Action expression: the human should clearly express the requested interaction or
↪  contact-ready pre-action state.
2. Target-object preservation: the target object should remain fixed in pose,
↪  location, scale, material, and appearance relative to the original scene.
3. Plausible body pose: the inserted human should have a physically plausible
↪  full-body pose, orientation, scale, and support.
4. Contact readiness: the relevant body parts should appear close to or in natural
↪  contact with the intended target object surfaces or floor support regions.

Intended output:
- Return the index or identifier of the best candidate and a concise reason for the
↪  selection.
```

A.1.4 Agentic Contact-Overlay System Prompt Π_{con}

```
Provided Images:

Reference Image: An image showing a human interacting with the target object. Use this
↪  image only to infer which human body parts are physically contacting the target
↪  object and which object-surface regions are being touched.

Canvas Image: An image of the same scene without the human. Use this image as the base
↪  image and as the authoritative reference for the target object's final position,
↪  pose, geometry, and boundaries.
```

```
Task Description:

Generate an edited copy of the Canvas Image with solid-color segmentation masks
↪ overlaid only on the target object surface regions that are contacted by the human
↪ in the Reference Image.

Infer the contacted region relative to the target object in the Reference Image, then
↪ place the corresponding contact mask on the matching object region in the Canvas
↪ Image.

Critical Base-Image Rule:
Use the Canvas Image as the only valid base image. Preserve the Canvas Image exactly
↪ and only add the specified solid-color contact masks. Do not use the Reference
↪ Image as the base. Do not redraw, regenerate, stylize, relight, crop, pad, resize,
↪ move, or otherwise alter the Canvas Image.

Analysis Requirements:

1. Contact Inference:
   Analyze the Reference Image to determine which specified human body parts are
   ↪ physically touching the target object.

2. Object-Region Identification:
   For each contacting body part, identify the specific region of the target object
   ↪ being touched, such as the tabletop front edge, tabletop side edge, tabletop
   ↪ upper surface, tabletop underside, table leg, chair seat, chair arm, cabinet
   ↪ handle, ladder rung, rail, shelf, handle, edge, or other relevant object part.

3. Object-Relative Localization:
   Map each inferred contact region onto the corresponding part of the target object
   ↪ in the Canvas Image. Use the Canvas Image object pose and boundaries as ground
   ↪ truth. Prioritize correct placement on the object in the Canvas Image over
   ↪ exact pixel alignment with the Reference Image.

4. Contact Footprint Generation:
   Create a localized contact mask only on the object surface where contact occurs.
   ↪ The mask should represent the apparent contact footprint on the object, not the
   ↪ full visible human body part.

Execution Requirements:

Base Image:
Return the Canvas Image with the contact masks overlaid on top of it.

Canvas Size:
Match the Canvas Image dimensions and coordinate system.

Mask Placement:
Place each mask on the corresponding target-object region in the Canvas Image, using
↪ object-relative correspondence rather than direct coordinate transfer.

Mask Shape:
Use precise, dense, solid-color segmentation masks shaped like the localized contact
↪ footprint on the object surface. The mask should reflect the approximate size,
↪ orientation, and extent of the contact implied by the touching body part.

Object-Surface Clipping:
All mask pixels must be clipped strictly to the target object boundary in the Canvas
↪ Image. No mask may extend onto the human body, floor, wall, background, or any
↪ non-target-object region.

Scene Preservation:
```

```
Use the Canvas Image as a fixed camera reference. Preserve the camera pose, field of
↪  view, perspective, resolution, aspect ratio, room geometry, lighting, shadows,
↪  textures, and all existing object identities.
Keep the target object in exactly the same pose, position, orientation, scale,
↪  material, and appearance as in the Canvas Image. Do not move, rotate, deform,
↪  lift, translate, resize, crop, pad, or reposition the target object or any other
↪  scene element in any way.
Only add the specified solid-color contact mask overlays. Leave every non-overlay
↪  pixel visually unchanged from the Canvas Image.

Colors:
Use only the exact specified solid colors for the contacting body-part masks. Do not
↪  use transparency, antialiasing gradients, shadows, blended colors, or pastel
↪  variants.

Restrictions:
Do not add the generated human body.
Do not copy or paste the full visible hand, foot, hip, arm, leg, or other body-part
↪  silhouette.
Do not transfer masks by raw Reference Image pixel coordinates.
Do not add text labels, arrows, legends, boxes, keypoints, outlines, or any other
↪  annotations.
Show only the original Canvas Image with solid localized contact masks overlaid on the
↪  target object.

Example Guidance:
If the Reference Image shows a person gripping a ladder, infer which rungs or rails
↪  the hands and feet contact. Then place solid masks on those same ladder regions in
↪  the Canvas Image, aligned to the ladder as it appears in the Canvas Image.
```

A.1.5 Contact VLM Evaluation System Prompt Π_{eval}

```
You are evaluating colored contact-region masks for a human-object interaction task.

You are given three images:

Image 1: Reference Image.
This image shows a human physically interacting with the target object. Use it to
↪  infer which body parts are touching which object surfaces.

Image 2: Original Canvas Image.
This image shows the same scene without the human. This is the authoritative geometry
↪  and base image.

Image 3: Current Composite Image.
This is the original Canvas image with colored contact masks pasted onto it. Evaluate
↪  only the colored masks.

Target object:
{target_object}

Color mapping:
{color_mapping}

Task:
Decide whether the colored masks in the Current Composite Image correctly mark the
↪  localized contact regions implied by the Reference Image.

Focus especially on location. For each mask, check whether it is on the correct object
↪  part and correct local surface.
```

```
Evaluate these possible errors:
1. Missing contact: a body part touches the target object in the Reference Image but
↪  its mask is absent.
2. Extra contact: a mask exists for a body part that does not touch the target object.
3. Wrong color: a contact uses the wrong color for the body part.
4. Wrong object surface: the mask is on the wrong object part, such as wrong rung,
↪  wrong rail, wrong edge, wrong handle, wrong shelf, wrong tabletop region, etc.
5. Shifted mask: the mask is too far left, right, up, or down from the correct contact
↪  location.
6. Size error: the mask is too large or too small for the local contact footprint.

For every problem, write a concrete correction instruction for the next ChatGPT
↪  image-generation prompt.

Good correction examples:
- "Keep the original canvas unchanged and move only the green right-foot mask upward
↪  onto the lower ladder rung."
- "The yellow left-hand mask is missing. Add a small yellow mask on the vertical side
↪  rail where the left hand contacts the ladder."
- "The blue foot mask is on the wrong rung. Move it down to the rung directly under
↪  the visible foot contact."
- "The red right-hand mask is too large. Keep its center but make it smaller and more
↪  focused."
- "This mask is correct. Keep it unchanged."

Bad correction examples:
- "The result is bad."
- "Fix the mask."
- "Try again."

Return strictly valid JSON only:

```json
{
  "done": false,
  "confidence": 0.0,
  "correction_instruction": "text"
}
```

Set "done": true only if all required contacts are present, there are no extra masks,
↪ each mask uses the correct color, each mask is on the correct target-object
↪ surface, and each mask has acceptable location and size.
```